

On the Prediction of Future Geomagnetic Activities Using Artificial Intelligence Techniques

Mirkan Y. Kalkan¹, Kıymet Yıldır², Dıaa E. Fawzy³

¹Physics, Institute of Graduate Studies, Ondokuz Mayıs University, Samsun, Türkiye

²Electrical and Electronics Engineering, Izmir University of Economics, Izmir, Türkiye

³Aerospace Engineering, Izmir University of Economics, Izmir, Türkiye

Abstract

The present study investigates machine-learning approaches for forecasting long-term geomagnetic activity using the planetary Ap index. The monthly AP data were smoothed with a 13-month trailing mean and modelled using Ridge Regression, Support Vector Regressor (SVR), Random Forest (RF), Extra Trees (ET), Natural Gradient Boosting (NGBoost), and Explainable Boosting Machine (EBM), Long Short-Term Memory (LSTM). The performance of the model was evaluated using a range of performance metrics, including the root mean square error (RMSE), the Pearson correlation coefficient (PCC), the standardised mean absolute percentage error (sMAPE), and the Nash-Sutcliffe efficiency (NSE). These metrics were used to assess the model's performance in relation to a persistence baseline. Ridge Regression demonstrated the most favourable one-month-ahead performance (RMSE = 0.258, PCC = 0.996, NSE = 0.992, sMAPE = 2.326) and exhibited a reduction in RMSE of approximately 11.8% relative to persistence. The findings indicate a gradual decline in smoothed geomagnetic activity from a maximum Ap of approximately 15.27 in January 2026 towards approximately 3–4 by 2030.

Motivation

1. Geomagnetic activity exerts a significant influence on a variety of critical infrastructure and systems, including satellite operations, navigation, communications, and technological infrastructure.
2. The planetary Ap index is a quantitative measure of global geomagnetic activity.
3. The utilization of data-driven methodologies has the potential to yield valuable insights pertaining to both short-term variability and the progression of activity throughout the solar cycle.

Data

Input: Monthly planetary Ap index

Historical training: 1933–2018

Out-of-sample validation: 2019–2025

Projection: 2026–2030

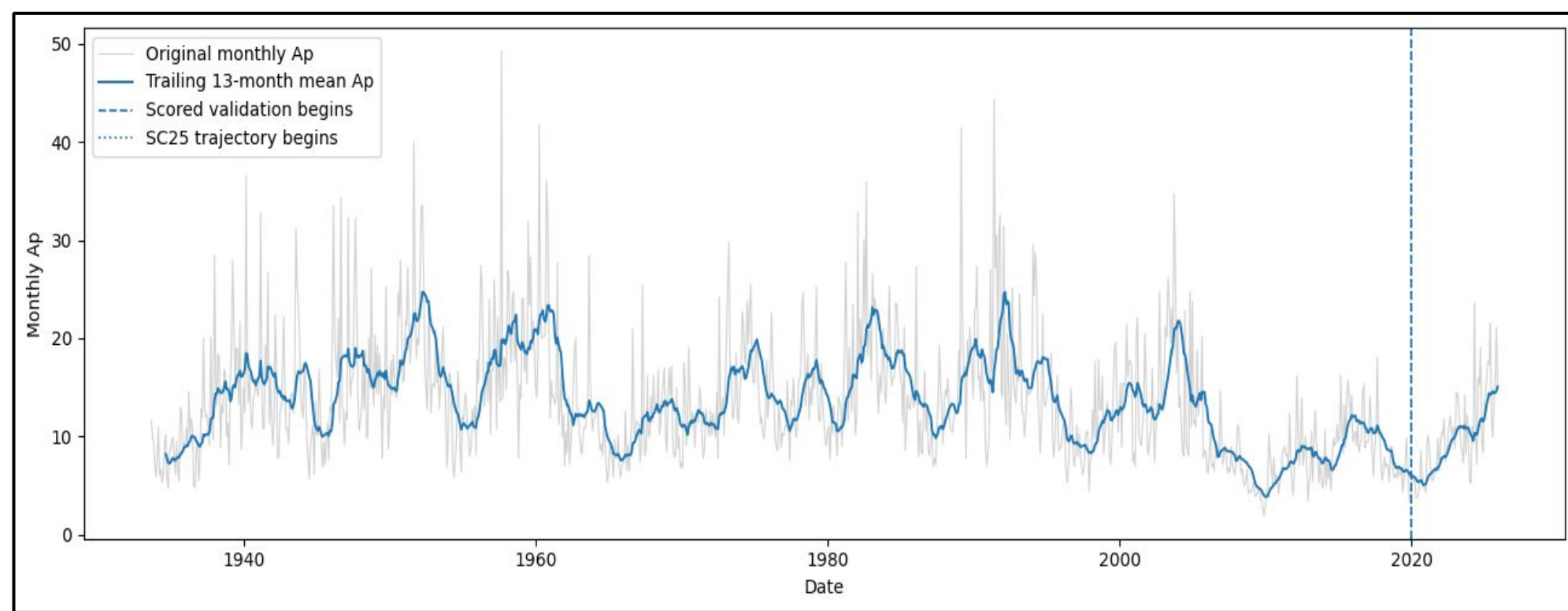
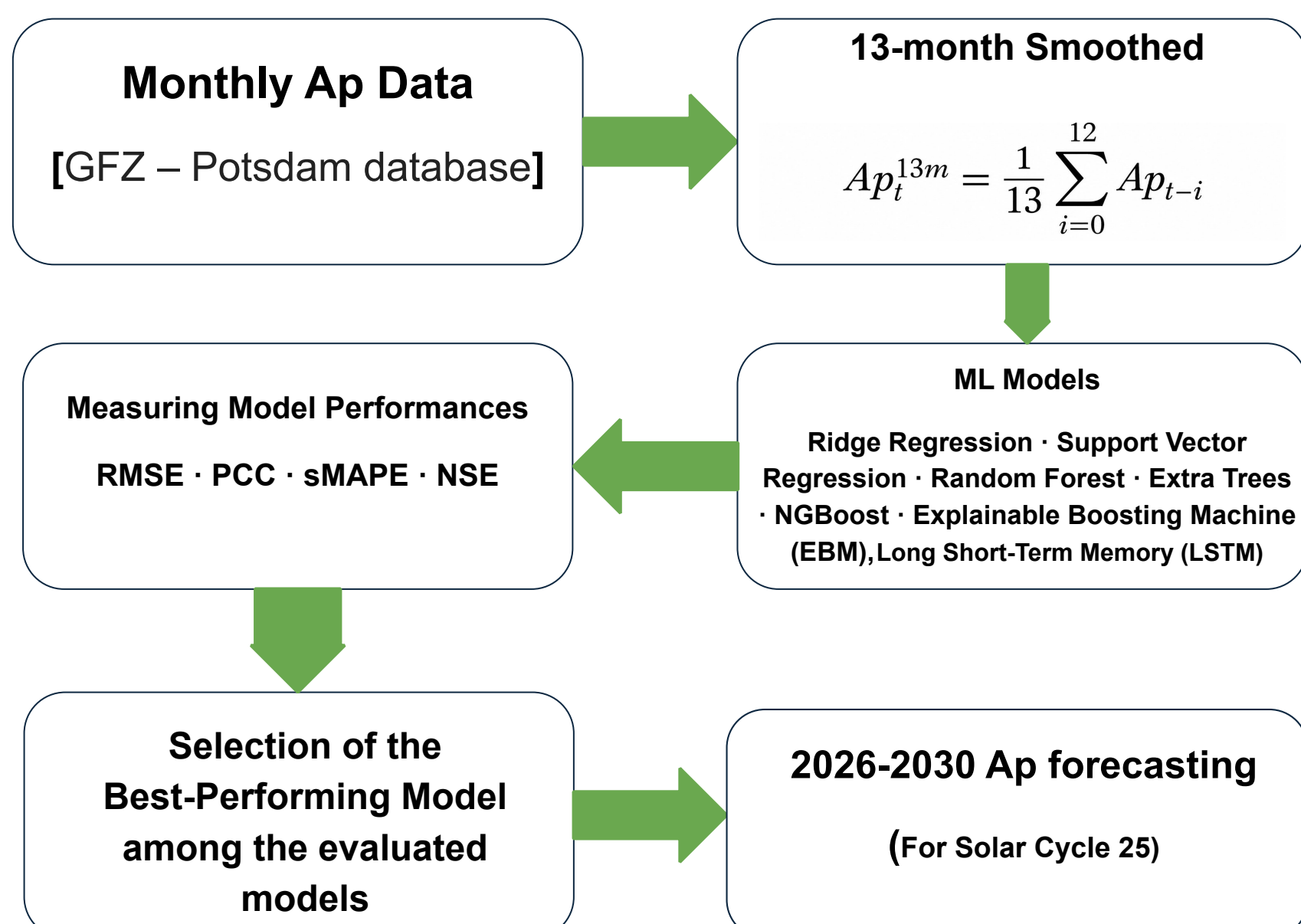


Figure 1. Observed monthly Ap and derived 13-month smoothed mean Ap data during the period 1933 to 2025

Methodology



- Monthly Ap values were smoothed using a 13-month trailing mean to emphasize solar-cycle-scale variability (Xu et al., 2008), while addressing the relatively limited use of multi-model machine learning for long-term Ap evolution compared with short-term ap forecasting (Mavromichalaki et al., 2024).
- The model set was selected to represent complementary linear, kernel-based, tree-ensemble, boosting, and neural-network learning strategies, following previous solar-cycle forecasting studies emphasizing model diversity, validation, and control of overfitting (Kalkan et al., 2023, 2026).

Table 1. Prediction results and performance metrics of 13-month smoothed Ap prediction models.

Model	RMSE	PCC	NSE	sMAPE
Ridge	0.258	0.996	0.992	2.326
SVR	1.252	0.981	0.818	13.012
RF	0.377	0.993	0.984	3.598
ET	0.391	0.991	0.982	3.653
NGBoost	0.299	0.995	0.990	2.718
EBM	0.285	0.996	0.991	2.450
LSTM	0.417	0.990	0.980	3.982

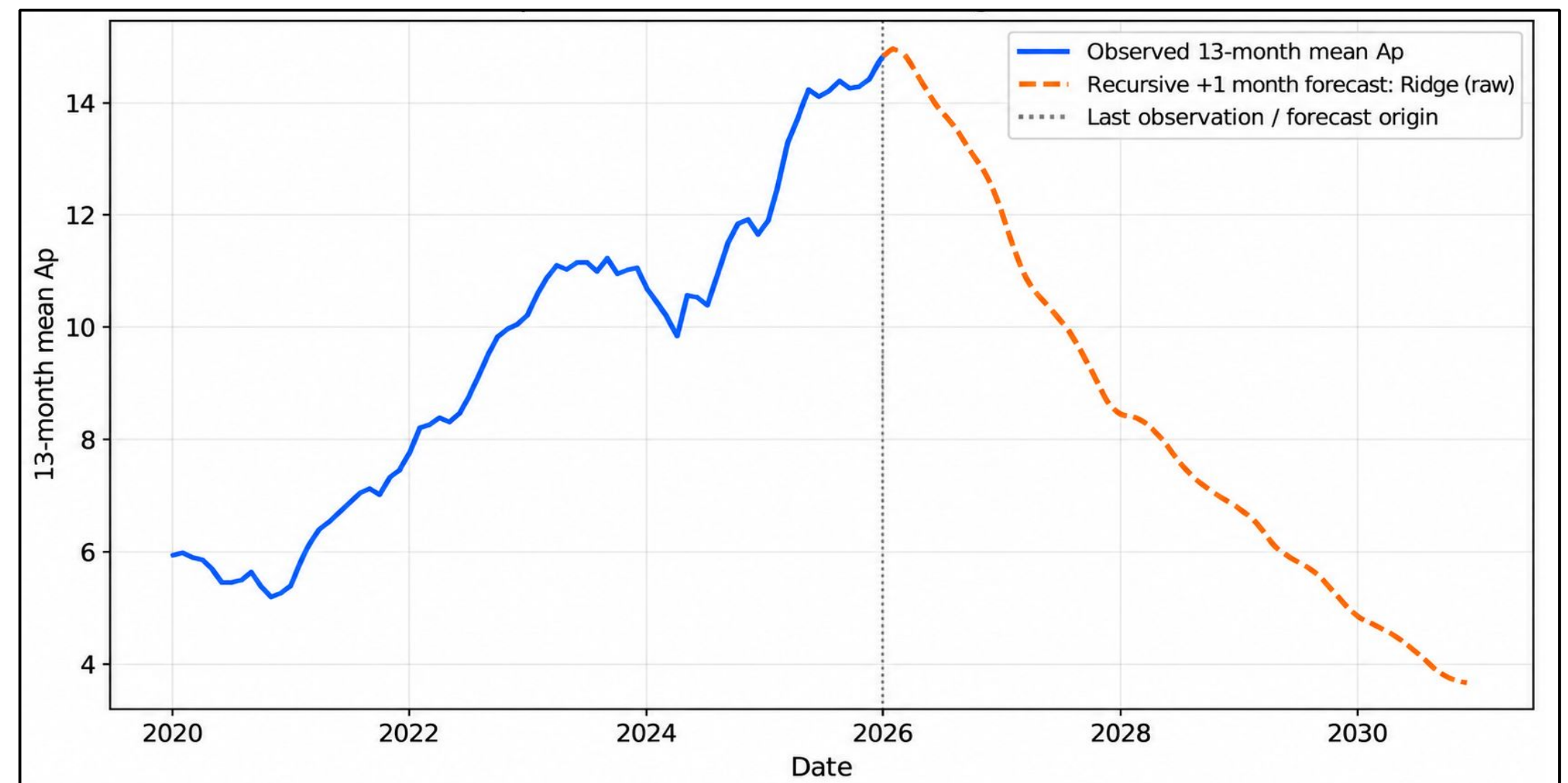


Figure 2. Observed and predicted 13-month mean Ap using Ridge Regression (2020–2030).

Year	Yearly-Mean Predicted Ap levels
2026	13.82
2027	9.72
2028	7.22
2029	5.42
2030	3.87

Table 2. Yearly-Mean Predicted Ap levels with Ridge Regression by years 2026 - 2030.

Conclusion

- Ridge Regression demonstrated the most favourable overall one-month-ahead performance among the evaluated models. Its superior performance is consistent with the smooth and strongly autocorrelated structure of the 13-month mean Ap series, where a regularized linear model can capture the dominant temporal evolution without introducing unnecessary nonlinear complexity.
- The Ridge projection attained its maximum smoothed Ap value of approximately 15.27 in January 2026, following which the recursive projection anticipates a gradual transition towards diminished geomagnetic activity, with a projected smoothed Ap level of approximately 3–4 by 2030.

References

- Ap data: GFZ – Potsdam database, <https://spaceweather.gfz.de/de/>
- Xu, T. et al. (2008), *Chinese Journal of Space Science*, 28.
- Mavromichalaki et al. (2024), *Atmosphere*, 15, 1073.
- Kalkan, M. Y. et al. (2023). *Monthly Notices of the Royal Astronomical Society*, 523(1), 1175–1181.
- Kalkan, M. Y. et al. (2026). *Publications f the Astronomical Society of the Pacific*, 138(4), 044502.